Introduction

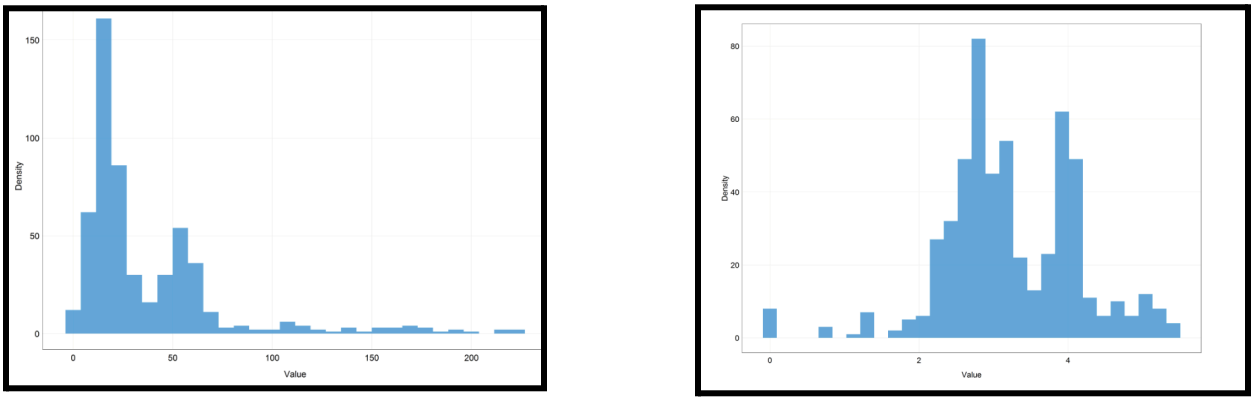
ScType is an algorithm that allows for quick cell type identification based on scRNA sequence data. ScType app is fast and accurate and is fully automated, additionally, ScType incorporates SNV calling to separate healthy and malignant cells. ScType was written in R and uses the Seurat R package along with manual cell type annotation to identify cells. ScType's accuracy can be attributed to the fact that gene markers are classified as positive or negative per cell.

Data Collection of ScType

Many models are only as good as the quality of its data. ScType takes data from many places with good practices to lead an accurate annotation tool. Data was taken from the CellMarker database, Human Protein Atlas, etc. and marked different markers as either positive or negative. The markers were collected from many different types of tissue and cell types across human and mouse datasets. This detailed extrapolation of markers allowed for greater accuracy. The markere database is regularly updated and open to community contributions. This allows for ScType to adapt as new discoveries are made.

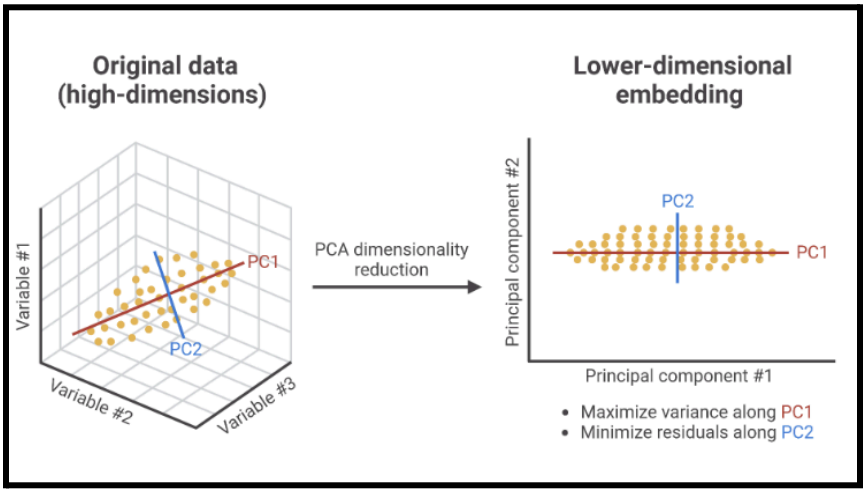
Preprocessing of ScType

ScType was written in R and uses the R Package Seurat. Seurat is a package built specifically for analysis, quality control, and manipulation of scRNA sequence data. ScType starts off with expecting normalized data. Specifically, data that has gone through the log1p transformation. Log1p transformation is used to help remove major outliers in a dataset, along with separating large clumps of data and bringing together sparse data.



[Original vs Normalized Data]

As seen from the image above, when data is normalized major outliers are not as prominent and the data is less skewed. This transformation is done by $X'_{g,n} = ln(1 + X_{g,n})$, where $X'_{g,n}$ is the matrix of genes and the number of single cells. After the data is logarithmized, the data is then scaled and centered, specifically with z-score normalization. For each gene in the data the transformation of $X''_{g,n} = \frac{X'_{g,n} - \mu_g}{\sigma_g}$, μ_g is the mean expression of g and σ_g is the standard deviation of g. Z-score normalization so that the data has a mean of 0 and a standard deviation of 1. Z-score normalization allows for multiple datasets to be compared to each other regardless of units as the new values represent z-scores which are irrespective of units. In the context of ScType z-score normalization ensures that genes contribute equally during the clustering. Without z-score normalization highly expressed genes would dominate variance and create bias during analysis later on. Finally dimensionality reduction is utilized before clustering. Specifically, Principal Component Analysis (PCA) is used to reduce dimensionality in clustering. This is denoted as $Z = W^T X$, where X is the scaled matrix, W^T is the matrix of the top eigenvectors of the covariance matrix , and Z represents the computed PCA scores. The covariance matrix essentially determines the behavior of two variables in respect to each other. The top eigenvectors(vectors that do not change direction during linear transformations) determine the maximum variance of the dataset. Eigenvectors are the principal components drawn and usually marked in the direction where the data has the most variance. Ensuring that data is preserved.



Original vs Lower-Dimension Data

The image above demonstrates how data is simplified due to PCA being applied. A 2D field was found inside of a 3D space, lower dimensional data simplifies datasets while preserving key information. PCA enables for easier data interpretation and exploration. PCA can also help data run faster. PCA is the final step in preprocessing for ScType.

Clustering of ScType

After the data is processed, using the louvain clustering algorithm from seurat communities are created from the data. Louvian clustering is a hierarchical clustering algorithm which creates groups inside of the dataset called “communities”. Communities represent groups of nodes that are more densely connected than other nodes from outside the group. These communities represent different ‘cells’ in this particular scenario. Now that communities are created these groups need to be annotated to their correct cell type. Louvian clustering is created from a K-nearest neighbours graph where the nodes are cells and edges are the similarity of PCA space. Euclidean distance is used as the default metric of K-nearest neighbors. Louvian clustering is expressed as $Q = \frac{1}{2m}\Sigma_{ij}[A_{ij} - \frac{k_i k_j}{2m}]\delta(c_i, c_j)$. The Louvain Algorithm aims to optimize modularity(Q) , the strength of divisions inside of a network. Higher Q indicates more defined communities inside of louvain clustering. The Kronecker delta function $\delta(c_i, c_j)$ is either 0 or 1, determined for if c_i and c_j are in the same community. The resolution value can also be altered to change the amount of communities created by the louvain clustering algorithm. Higher resolution leads to fewer and larger communities while the opposite can be said for lower resolutions. The output of louvain clustering is a set of clusters, and each cluster represents a cell population.

Manual Marker Annotation of ScType

What makes ScType so accurate is the ability to account for positive and negative markers, this is done by having a marker dictionary with a list of cells with positive and negative genes for each cell type. For example T cells might have CD3D and CD3E as positive markers and CD14 as a negative marker. For each community created by louvain clustering, a scoring algorithm is run on the cluster for each cell type. This means that for each cluster, a score is assigned for each possible cell type that can be assigned to a cluster. The gene type with the highest score is considered to be the gene for the cluster. First, the markers are filtered to be the ones specifically for the gene. A specificity weight is applied which is between 0 and 1 and helps favor/punish certain markers more heavily. The marker enrichment score is calculated per cell, each marker enrichment score inside of a cluster is added up. The marker enrichment score formula is $x'_c = \frac{\sum_{i=1}^{j} x'_i}{\sqrt{j}} - \frac{\sum_{i=1}^{l} x'_i}{\sqrt{l}}$, where j and l are the number of positive and negative markers respectively. $x'_i \; and \; x'_j$ are the transformed values of the positive and negative markers respectively. The summation of x'_c across the entire cluster for a specific cell type gives the score from ScType. The negative summation is subtracted from the positive summation to lower the overall score of a gene t in a given cluster c. Essentially, this manual annotation prioritizes expression of positive markers and punishes expression of negative markers. Additionally, markers have specificity weights to prioritize/punish a marker more. Finally, after all the scores are created for each gene in each cluster. The highest score for each gene in a given cluster is deemed to be the cell type of the cluster. A threshold can also be applied so that only confident assignments are kept instead of low scores. Thus avoided false positive assignments, and being marked as “unknown” instead.